

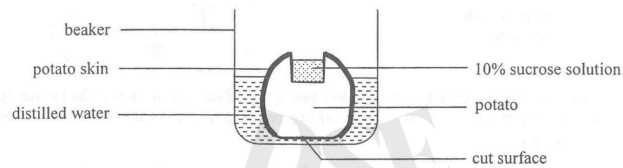
1. 2013Q5

Which of the following descriptions of the function of membrane proteins is **incorrect**?

- A. Some membrane proteins serve as antigens for cell recognition.
- B. Some membrane proteins serve as enzymes for cellular metabolism.
- C. Some membrane proteins act as barriers which prevent the entry of some substances.
- D. Some membrane proteins act as carriers which transport some substances across the membrane.

2. 2013Q23

Questions 23 to 25 refer to the diagram below, which shows an experiment on osmosis using a potato tuber. A washed potato was cut to form a base. After that, a cavity was made and a 10% sucrose solution was added into the cavity. The whole potato was then placed into a beaker containing some distilled water. After 1 day, the level of sucrose solution rose.



The level of sucrose solution inside the cavity rose because osmosis has taken place, which means that

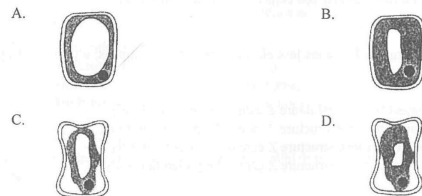
- A. water was drawn mainly from the cells in contact with the sucrose solution.
- B. water was initially drawn from the cells in contact with the distilled water.
- C. water moved from the distilled water along the cell wall of the potato cells to the cavity.
- D. water moved from the distilled water through the cell membrane of the potato cells to the cavity.

3. 2013Q24

The diagram below shows the appearance of a cell in contact with distilled water in the beaker at the beginning of the experiment:



Which of the following diagrams correctly shows the appearance of the same cell at the end of the experiment?



4. 2013Q25

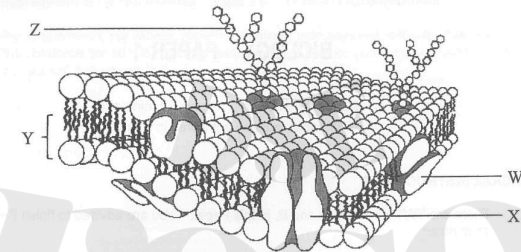
Which of the following treatments will lead to a higher final level of sucrose solution after 1 day?

- (1) using 5% sucrose solution instead of 10% sucrose solution
- (2) using 20% sucrose solution instead of 10% sucrose solution
- (3) peeling off all the potato skin instead of just cutting the bottom of the potato

- A. (1) only
- B. (2) only
- C. (1) and (3) only
- D. (2) and (3) only

5. 2016Q1

Question 1 refers to the schematic diagram below, which shows the structure of a cell membrane:

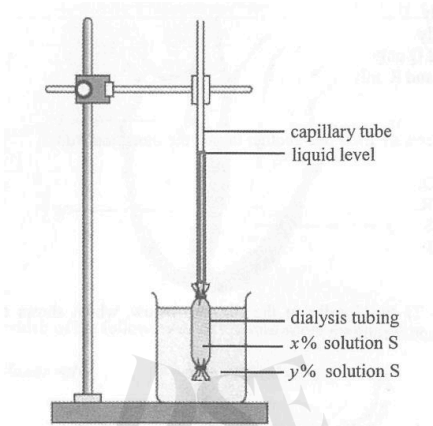


Which of the following molecules contributes to the fluidity of the cell membrane?

- A. W
- B. X
- C. Y
- D. Z

6. 2016Q24

Questions 24 and 25 refer to the following diagram below, which shows an experimental set-up for studying osmosis:



- After three hours, the liquid level has risen. What can be deduced from this result?
- (1) Solution S diffuses into the dialysis tubing.
 - (2) Value of x is larger than that of y.
 - (3) Solute of solution S cannot pass through the dialysis tubing
- A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

7. 2016Q25
- Which of the following modifications will result in a faster rise in the liquid level?
- A. use a smaller volume of x% solution S
 - B. use a larger volume of y% solution S
 - C. use a shorter capillary tube
 - D. use a longer piece of dialysis tubing

8. 2019Q8

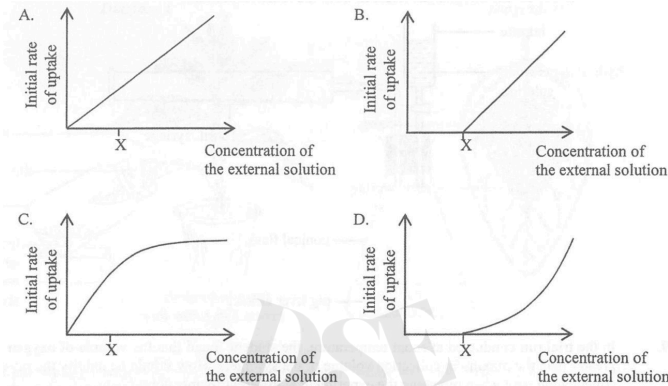
According to the requirement of various methods of transport across the cell membrane, which of the following combinations is correct?

Requirements		
Energy input	Membrane protein	Concentration gradient
A. phagocytosis	active transport	osmosis
B. diffusion	osmosis	active transport
C. active transport	phagocytosis	phagocytosis
D. osmosis	diffusion	diffusion

9. 2019Q11

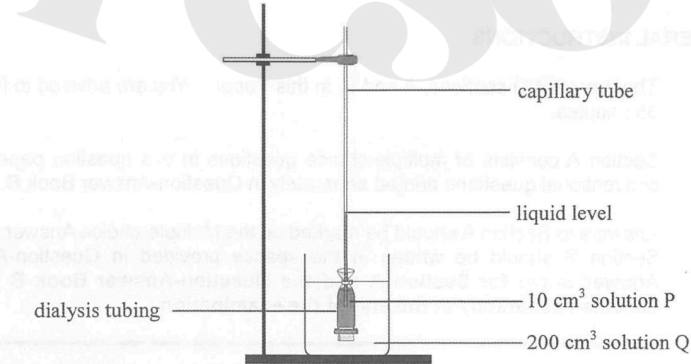
Solutions of different concentrations of a solute with a small molecular size were prepared and some plant cells were immersed in each of the solutions. Which of the following graphs shows the initial rate of uptake of the solute by means of diffusion?

(Note: X is the concentration of the solute inside the plant cells)



10. 2020Q2

Questions 2 and 3 refer to the diagram below, which shows a set-up for investigating the permeability of dialysis tubing. 10 cm³ of solution P was added to the dialysis tubing, with one end tied and the other end connected to a capillary tube. The dialysis tubing was then placed in a beaker with 200 cm³ of solution Q. When solutions P and Q are mixed, a blue colour will be observed.



- At the end of the investigation, the liquid level inside the capillary tube had risen and only the solution inside the dialysis tubing became blue. Which of the following conclusions can be drawn from the results?
- (1) Solute of solution P can pass through the dialysis tubing
 - (2) Solute of solution Q can pass through the dialysis tubing
 - (3) There is a net movement of water into the dialysis tubing
- A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

11. 2020Q3

If the investigation is repeated with water instead of solution Q, which of the following results will be obtained?

- A. The liquid level in the capillary will not rise at all.
- B. The liquid level in the capillary will rise to a lower level.
- C. The liquid level in the capillary will rise to a higher level.
- D. The liquid level in the capillary will rise to the same level.

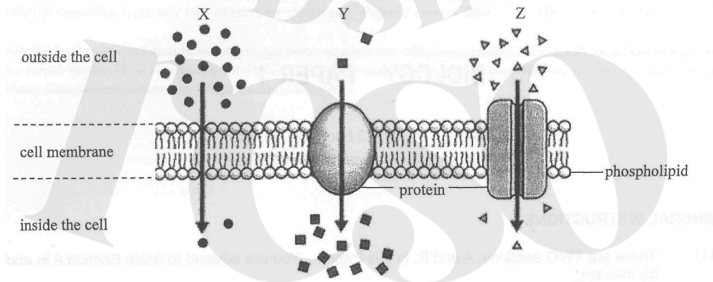
12. 2021Q8

Which of the following combinations correctly lists the requirements of both osmosis and diffusion?

	Concentration gradient	Energy supply	Selectively permeable membrane
A.	Yes	No	Yes
B.	Yes	No	No
C.	No	Yes	Yes
D.	No	Yes	No

13. 2022Q1

The diagram below shows three different ways by which substances pass through a cell membrane:

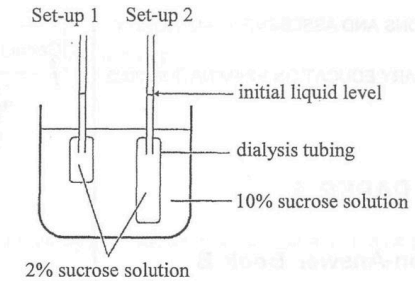


Which of the following combinations correctly matches the substances with their corresponding ways of passing through the cell membrane?

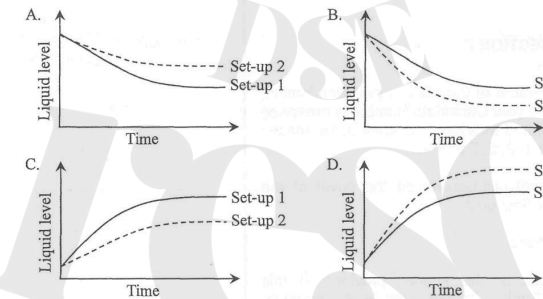
	X	Y	Z
A.	oxygen	glucose	water
B.	glucose	water	oxygen
C.	water	oxygen	glucose
D.	water	glucose	oxygen

14. 2022Q35

Questions 35 and 36 refer to the diagram below, which shows two set-ups used for investigating osmosis. Two dialysis tubings of different lengths were filled with 2% sucrose solution and then submerged in 10% sucrose solution. All solutions were kept at 25°C.

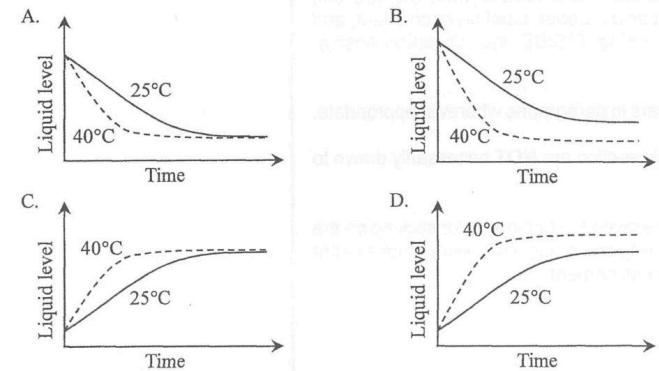


Which of the following graphs correctly shows the change in the liquid levels of set-ups 1 and 2?



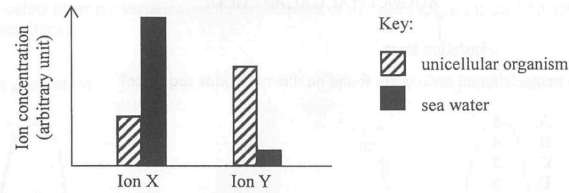
15. 2022Q36

If the experiment was repeated with the same setting except that the solutions were kept at 40°C, which of the following graphs correctly shows the change in the liquid levels of set-up 1 at different temperatures?



16. 2023Q31

The graph below shows the concentration of ions in a unicellular organism and the sea water surrounding it:

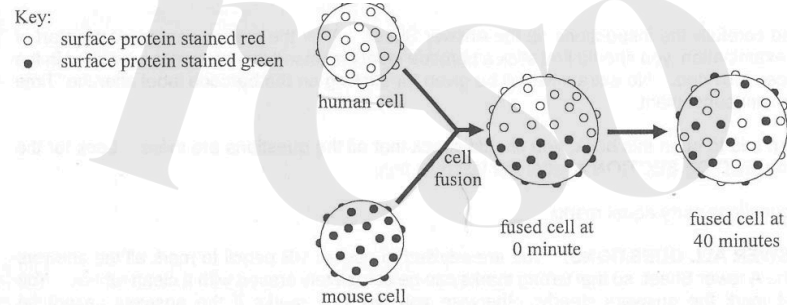


Which of the following is most likely to occur in order to maintain the conditions shown in the above graph?

- Ion X moves out of the organism by diffusion.
- Ion X moves into the organism by active transport.
- Ion Y moves into the organism by active transport.
- Ion Y moves out of the organism by active transport.

17. 2024Q3

The schematic diagram below shows an investigation into cell membranes. The cell surface proteins of a human cell were labelled with a red dye while those of a mouse cell were labelled with a green dye. The two cells were then fused. The distribution of the cell surface proteins at 0 minute and 40 minutes are shown below:



Which of the following properties of the cell membrane can be used to explain the change in the distribution of the surface proteins of the fused cell shown in this investigation?

- A cell membrane is fluid in nature.
- A cell membrane is differentially permeable.
- A cell membrane is an asymmetric structure.

- (1) only
- (2) only
- (1) and (3) only
- (2) and (3) only

LQ

1. 2012Q8a

The photograph below shows the appearances of some kernels of a corn:



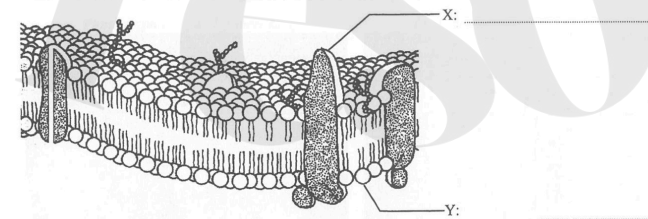
Kernel	Appearance
A	Purple and smooth
B	Purple and wrinkled
C	Yellow and smooth
D	Yellow and wrinkled

The purple colour is produced by a pigmented layer within the kernels. If the layer is not pigmented, the yellow colour of the inner tissue becomes visible. Whether the kernel is smooth or wrinkled is due to the type of food stored inside it. Smooth kernels (starchy corn) store starch while wrinkled kernels (sweet corn) store soluble sugars. The surface of the sweet corn becomes wrinkled when the corn dries up.

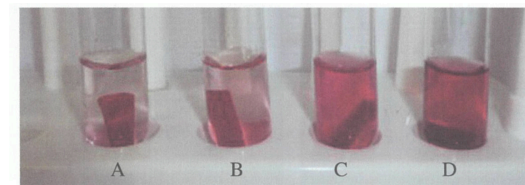
- With reference to osmosis, explain why the kernels of sweet corn become wrinkled when they dry up but the kernels of starchy corn remain smooth. (4 marks)

2. 2014Q7

- The following is a schematic diagram of a cell membrane. In the spaces provided, label membrane components X and Y. (2 marks)



- The vacuoles of beetroot cells contain a red pigment which will be released from the cells if the cell membrane and vacuole membrane are damaged. In an investigation, Gary placed identical cylinders of beetroot tissues into four test tubes. Each tube contained the same volume of alcohol at different concentrations. The following photograph shows the appearance of the solutions bathing the beetroot cylinders after 30 minutes:



- From the result of the above investigation, deduce which test tube contained the highest concentration of the alcohol (4 marks)
- After three hours, Gary found that the colour intensity of the solutions of all the test tubes became the same. Suggest an explanation for this. (2 marks)

3. 2015Q6

The table below lists some historical developments about discovery of the structure of cell membrane:

Year	Scientists	Historical events
1895	Overton	Discovered that lipid-soluble substances could penetrate cells easily
1917	Langmuir	Discovered that the major component of cell membrane exhibited both water-loving and water-hating properties
1925	Gorter & Grendal	Extracted lipids from the cell membrane of red blood cells and spread the lipids in a single layer on a water surface; found that the area of the layer was double the surface area of the cell membrane
1972	Singer & Nicolson	Proposed the Fluid Mosaic Model to explain the structure of cell membranes

- a) What is the major component noted by Overton and Langmuir? (1 mark)
- b) Gorter and Grendal proposed that the major component identified in (a) existed as a bilayer (Bilayer Model). With reference to the observation of Langmuir, suggest how this major component is oriented and arranged in the cell membrane. Explain your answer. (3 marks)
- c) i) The Bilayer Model proposed by Gorter and Grendal did not mention another major component of the cell membrane. What is this component? (1 mark)
- ii) With reference to the Fluid Mosaic Model, briefly describe the orientation of this component in the cell membrane. (2 marks)
- d) Models are often used by scientists to explain their findings. Complete the following table to elaborate on the aspects of the nature of science involved in the use of scientific models. (2 marks)

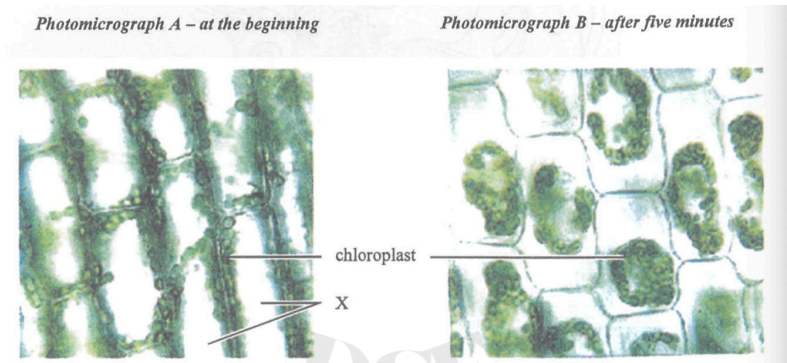
Nature of science	Elaboration
Science is evidence based	
	Models are used to stimulate an invisible structure or illustrate a theory.

4. 2017Q2

Denise ate a piece of pineapple preserved in a sugar solution and noticed that it was softer than fresh pineapple. Explain this phenomenon. (3 marks)

5. 2018Q2

The leaf of an aquatic plant was placed in a concentrated sucrose solution and observed under a light microscope. Photomicrographs A and B show the appearance of the cells at the beginning of the experiment and after five minutes respectively:



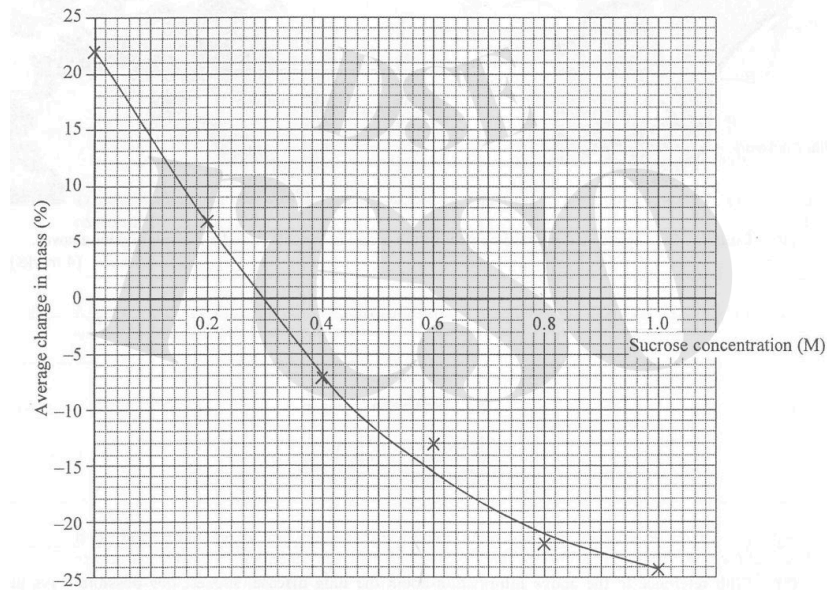
- a) X is an organelle which is invisible without staining. Name this organelle. (1 mark)
- b) Comparing the two photomicrographs, state **two** observable changes in the appearance of the cells after five minutes. (2 marks)
- c) Explain how the observable changes stated in (b) are brought about. (2 marks)

6. 2019Q6a, b, c

Johnny conducted an experiment to determine the water potential of potato tuber cells. He measured the masses of fresh potato cylinders before and after immersing them in sucrose solutions at different concentrations (0M, 0.2M, 0.4M, 0.6M, 0.8M, and 1.0M). Below shows the major steps in the experiment:

- Step 1: Cut potato tubers into cylinders
- Step 2: Blot dry the surface of the potato cylinders
- Step 3: Weigh the potato cylinders (initial mass)
- Step 4: Immerse three potato cylinders in each concentration of sucrose solution for two hours
- Step 5: Remove and blot dry the surface of the potato cylinders
- Step 6: Reweigh the potato cylinders (final mass)
- Step 7: Calculate the average percentage change in the masses of the potato cylinders in each solution.

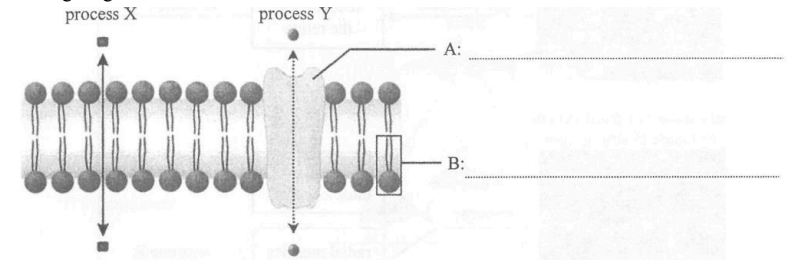
The results are shown in the graph below:



- a) With reference to the graph, which sucrose solution concentration has the same water potential as the potato cells? Explain your answer. (3 marks)
- b) If Johnny skipped step 2 by mistake for all samples, how would this affect the curve and the deduced value of the concentration of the sucrose solution in (a)? Sketch a curve on the graph to show the effect. (1 mark)
- c) In terms of experimental design, what is the importance of putting three potato cylinders in each concentration of sucrose solution? (1 mark)

7. 2020Q4a

The following diagram shows a cell membrane and the movement of some substances across it:



- a) Label structures A and B.

(2 marks)

Ans

2013Q5: C (41%)	2013Q23: D (48%)	2013Q24: A (65%)	2013Q25: B (24%)
2016Q1: C (85%)	2016Q24: C (36%)	2016Q25: D (58%)	2019Q8: A (73%)
2019Q11: B (23%)	2020Q2: C (85%)	2020Q3: C (71%)	2021Q8: B (38%)
2022Q1: A (56%)	2022Q35: B (59%)	2022Q36: A (63%)	2023Q31: C (63%)
2024Q3: A (57%)			

2012Q8a

- 8 (a) • soluble sugars lower the water potential of the kernels (1) while
 • starch, being insoluble in water, does not affect the water potential of the kernels (1)
 • during development, kernels of sweet corn draw in a larger amount of / more water by osmosis than those of starch kernels (1)
 • later on, when the kernels dry up, kernels of sweet corn shrinks as a result of losing a large amount of water (1)
 • as the outer layer of the kernels is not elastic (1), thus the kernels of sweet corns become wrinkled (max. 4)

2014Q7

- 7 (a) X: protein (1)
 Y: phospholipid (1) (2)
- (b) (i) • tube D should have the highest concentration of alcohol (1)
 • this is because the phospholipids of the membrane dissolve in alcohol (1)
 • the cell membrane and vacuole membrane of the beetroot tissue bathing in the test tube with the highest concentration of alcohol would be most damaged (1)
 • the amount of pigment released to the bathing solution would be the highest, as indicated by the highest colour intensity (1) (4)
- (ii) • when the cell membrane and vacuole membrane are damaged, the pigment leaks out of the vacuole by diffusion (1)
 • as time passes, it allows the diffusion of the red pigment in all 4 tubes to reach an equilibrium state at which the same concentration of red pigment are found in the bathing solutions / the bathing solution contains the same amount of red pigment in all 4 tubes, i.e. same colour intensities (1) (2)

8 marks

2015Q6

- 6 (a) phospholipids (1) (1)
- (b) • the water-loving parts of phospholipids point outward while the water-hating parts point inward (1)
 • the medium inside and the medium outside the cell are aqueous solutions (1)
 • hence, only the water-loving part of the phospholipids is in contact with the cell sap and extracellular fluid (1) (3)
- (c) (i) protein (1) (1)
- (ii) • proteins are interspersed in the bilayer (1)
 • in an asymmetric manner / a mosaic pattern (1) (2)

(d)

Nature of science	Elaboration	
-	Scientific models / theory are built based on experimental findings / observations (1)	(2)
Doing science requires imaginations / creativity (1)	-	

9 marks

2017Q2

- 2 • sugar solution has a lower water potential than the cell content of fresh pineapple (1)
 • as a result, water moves out of pineapple cells to the sugar solution by osmosis (1)
 • pineapple cells become flaccid (1) and fail to press against one another hence, the texture of pineapple preserved in sugar solution is softer than that of fresh pineapple (1) (3)

3 marks

2018Q2

- 2 (a) • vacuole (1) (1)
- (b) Any **two** of the following:
 • the cell membrane / cytoplasm of the leaf cells has detached from the cell wall / the leaf cell is plasmolyzed (1)
 • chloroplasts condense to the centre of the cell (1)
 • the vacuole / X has shrunk (1) (2)
- (c) • sucrose solution has a lower water potential than the cell content (1)
 • there is a net movement of water from the cell content to the bathing solution by osmosis (1) (2)

5 marks

2019Q6a, b, c

- 6 (a) • 0.3M sucrose solution (1)
 • there was no change of mass at this concentration (1)
 • this shows that there was no net movement of water in or out of the cells (1)
 i.e. the bathing solution had the same water potential as the potato cells (3)
- (b) • the curve would shift downward / to the left (1) (1)
- (c) • this can increase the reliability of the results by minimizing the individual differences of the cylinders (1) (1)

2020Q4a

- 4 (a) A: protein (1)
 B: phospholipid (1) (2)